



GE VERNOVA

ADMS

16.25.1

DNOM2HDB Installation Guide

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Change History

Date	Change Description
Jan 26, 2023	ADMS 3.16: • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Mar 25, 2024	ADMS 16.7.0: • Updated document format and styling.

Contents

1. Overview	4
1.1. Software Organization	4
1.2. Prerequisites	4
1.3. Installation Process	4
2. Software Installation	5
2.1. Installing DNOM2HDB	5
2.2. Building DNOM2HDB Individually	6
2.2.1. Loadhab	6
2.2.2. Clone	6
2.2.3. Build	7
2.3. Validating/Verifying the DNOM2HDB Installation	7
3. System Configuration	8
3.1. PERMIT	8
3.1.1. Defining the DNOM2HDB Application	8
3.1.2. Defining the DNAFAPP Application	12
3.2. PROCMAN	17
3.2.1. PROCMAN Model Changes	17
3.3. Configuration Control	19
3.4. Displays	22
3.5. Menus	22
3.6. Savecases	24
3.7. DMS Service Module	26
3.8. Troubleshooting	27
3.8.1. Enabling Detailed Logging	27
3.8.2. Running DNOM2HDB without PROCMAN	29

1. Overview

This chapter provides an overview of the installation process and describes the conditions that must be met prior to installing DNOM2HDB.

1.1. Software Organization

The software that is delivered with DNOM2HDB consists of:

- Habitat codeset: DNOM2HDB

1.2. Prerequisites

DNOM2HDB can be installed on Habitat , Platform (or Scada), and ADMS for Microsoft Windows and Red Hat Enterprise Linux operating systems.

1.3. Installation Process

1. Install the DNOM2HDB files using the `ADMS_dnom2hdb_X64.msi` kit.
2. Manually build the DNOM2HDB files.

2. Software Installation

This chapter describes the DNOM2HDB installation process.

2.1. Installing DNOM2HDB

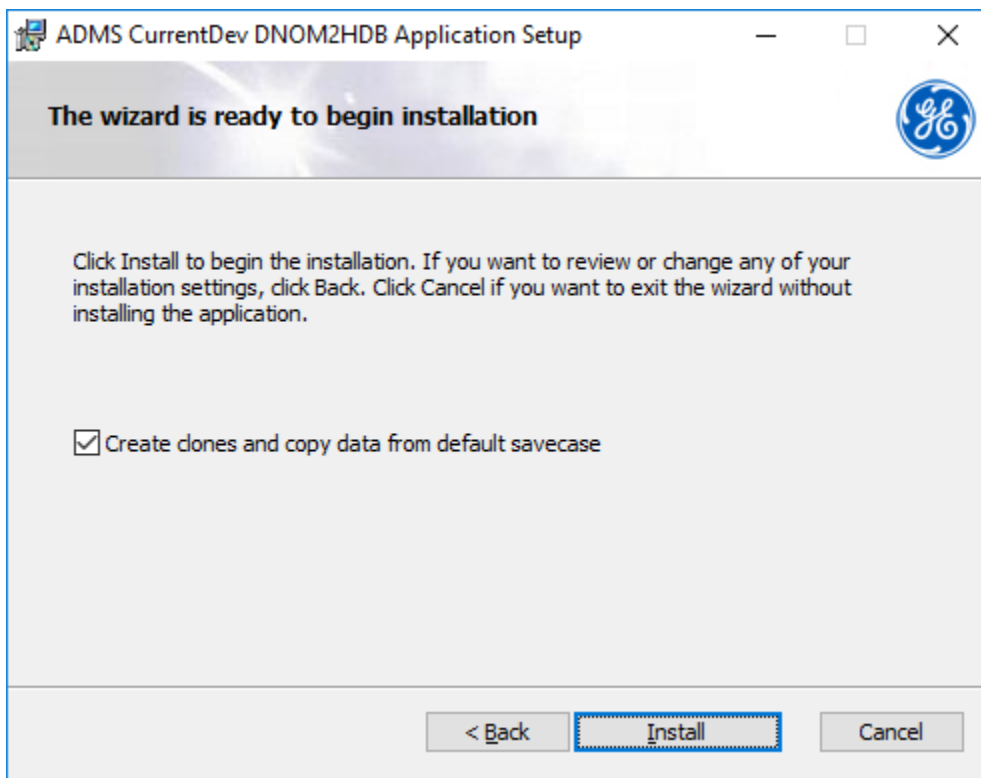
This section describes how to install the DNOM2HDB files using the ADMS_dnom2hdb_X64.msi kit.

Caution

The DNOM2HDB installation stops all Habitat applications that are running on the computer.

To install the DNOM2HDB files:

1. Run the ADMS_dnom2hdb_X64.msi installer.
2. Select the Create clones and copy data from default savecase option and then click Install.



3. Wait until the installation completes and then click Finish.

During the installation, DNOM2HDB files are copied to the following directories:

- D:\Eterra\habsrc06\habuser\special\Dnom2hdb\
- D:\Eterra\habweb06\displays\dnom2hdb\
- D:\Eterra\habdata06\appdefs\

- D:\Eterra\habdata06\cdbroot\clones\
- D:\Eterra\habdata06\cdbroot\dictionary
- D:\Eterra\habdata06\savecases\
- D:\Eterra\habdata06\uidbroot\ddls\
- D:\Eterra\habitat06\habuser\bin\

2.2. Building DNOM2HDB Individually

This section describes how to build a DNOM2HDB application individually (that is, outside of a system build).

Note

Do not perform the procedure below if you installed DNOM2HDB using the ADMS_dnom2hdb_X64.msi installer.

2.2.1. Loadhab

At the command prompt, go to the DNOM2HDB folder (by default, D:\Eterra \habsrc06 \habuser \special \Dnom2hdb) and load the Habitat files using the command:

```
loadhab *
```

2.2.2. Clone

At the command prompt, go to the DNOM2HDB folder (by default, D:\Eterra\habsrc06\habuser\special\Dnom2hdb).

- For a standalone system (for example, operator simulator or study environment), create the clone using the command:

```
hdbcloner -c create_clone -a _dnom2hdb_ -f _fam_
```

- For a dual redundant system (MRS is used), create the clone using the command:

```
hdbcloner -c create_clone -a _dnom2hdb_ -f _ems_ -replicate
```

Note

The command "hdbcloner -replace" allows you to update a clone structure and keep the current data.

2.2.3. Build

Build the DNOM2HDB codeset, and verify that a `dnom2hdb.exe` file is created in the `habitatxx/habuser/bin` folder with a date that corresponds to the time that the BUILD command was issued.

Note

If the Windows build fails because it cannot find some header files, verify that `habitat_environment_habdev_net_template.bat` sets up proper INCLUDE paths for the SDK kit. It is possible to manually set paths for VS2008: Tools/Projects/Projects and Solutions/VC++ Directories.

For more information, refer to the *ESCATOOLS User Guide*.

2.3. Validating/Verifying the DNOM2HDB Installation

Verify that the `dnom2hdb.exe` application can be launched from the Windows or Linux command prompt.

At the Windows command prompt, type:

```
> context dnom2hdb ems
> dnom2hdb
```

At the Linux command prompt, type:

```
% context dnom2hdb ems
% $HABUSER_BINDIR\dnom2hdb
```

Note

Habitat must be running to start `dnom2hdb.exe`.

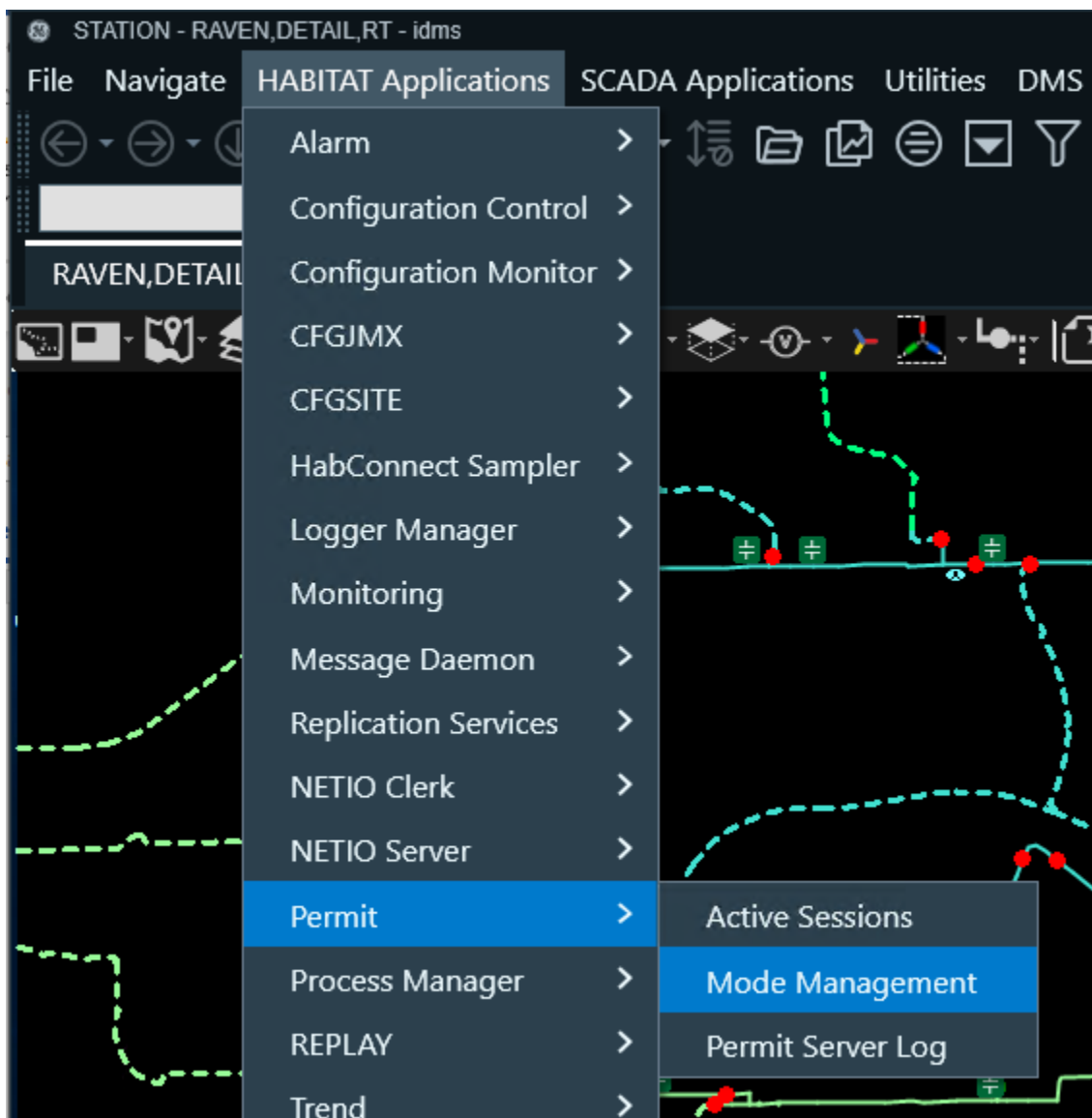
3. System Configuration

This chapter discusses some specific configurations that are required after a DNOM2HDB installation. For more information about any required steps, refer to the Habitat documentation.

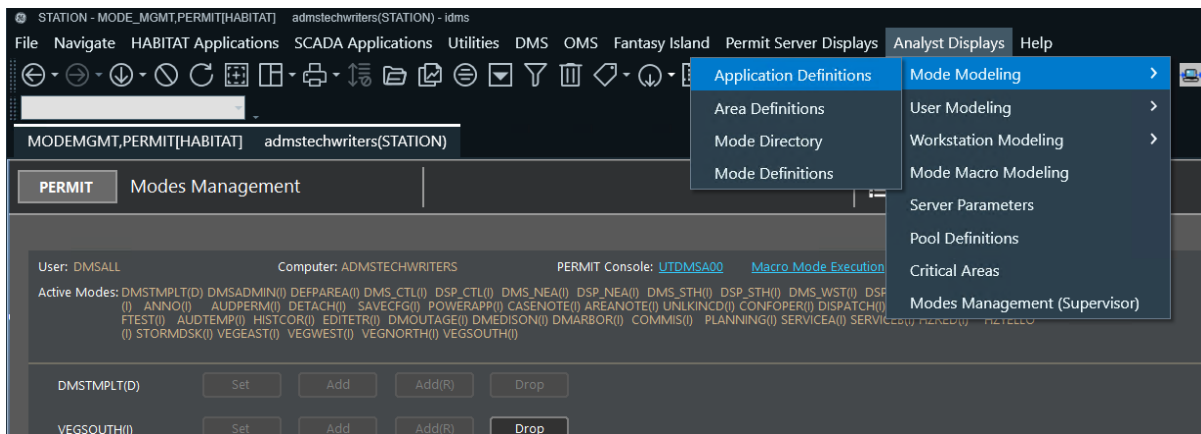
3.1. PERMIT

3.1.1. Defining the DNOM2HDB Application

1. In the Browser client, click HABITAT Applications > Permit > Mode Management.

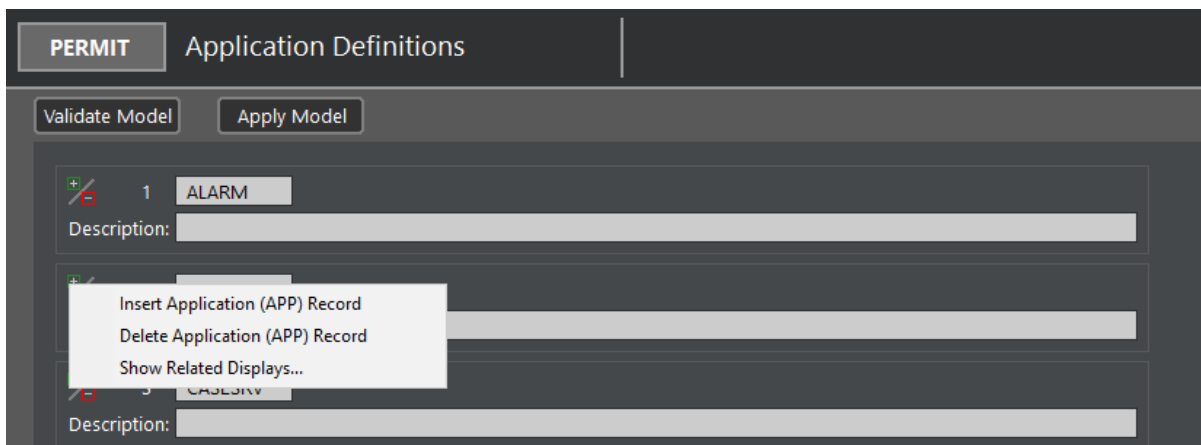


2. On the Analyst Displays menu, click Mode Modeling > Application Definitions.



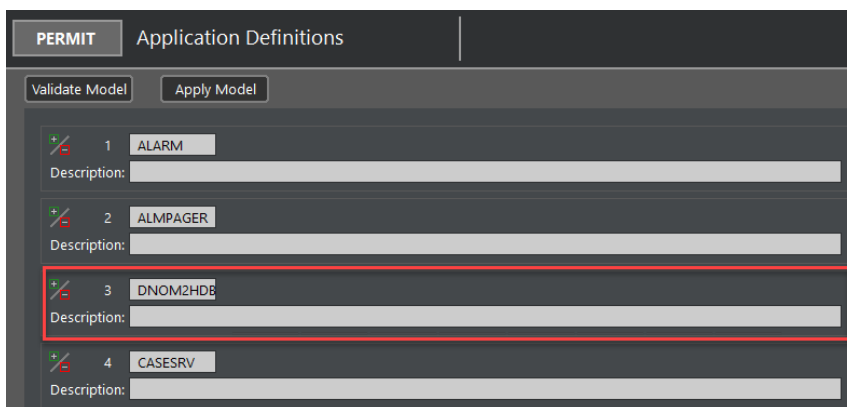
This opens the Application Definitions display.

- On the Application Definitions display, right-click the Insert or Delete Application Record field and select Insert Application (APP) Record.



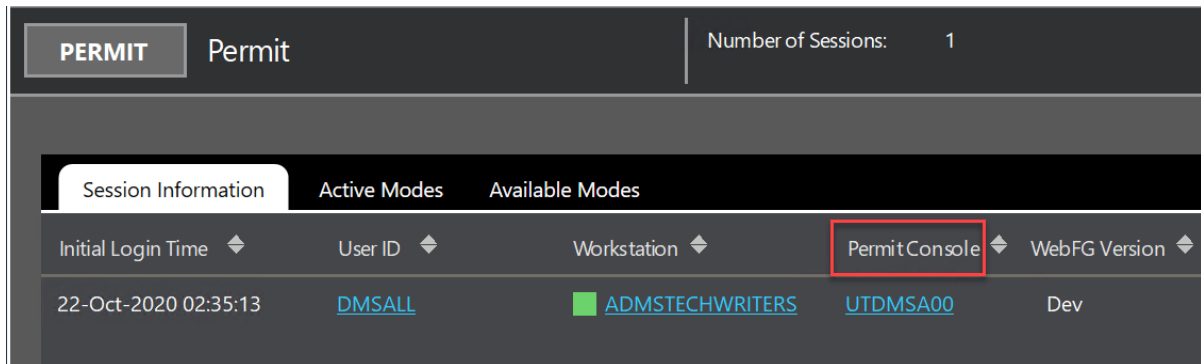
A new application record is added.

- Enter the DNOM2HDB application name and press Enter.

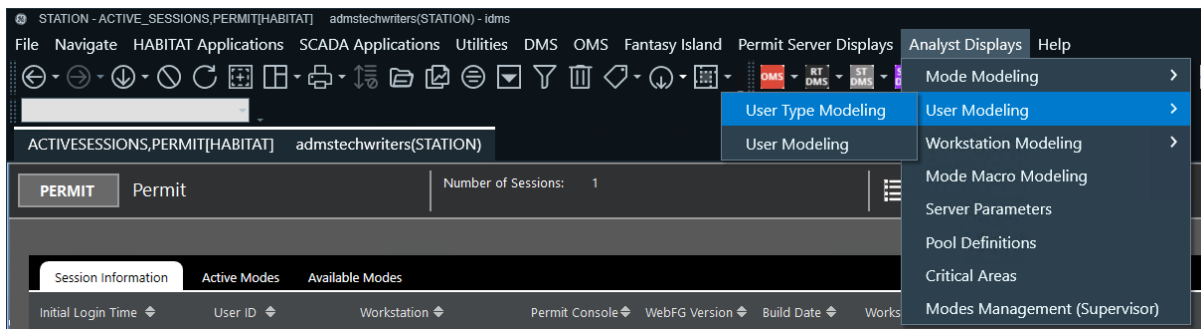


- Open the Permit display (HABITAT Applications menu > Permit > Active Sessions).
- On the Permit display, identify your user type on the Permit Console.

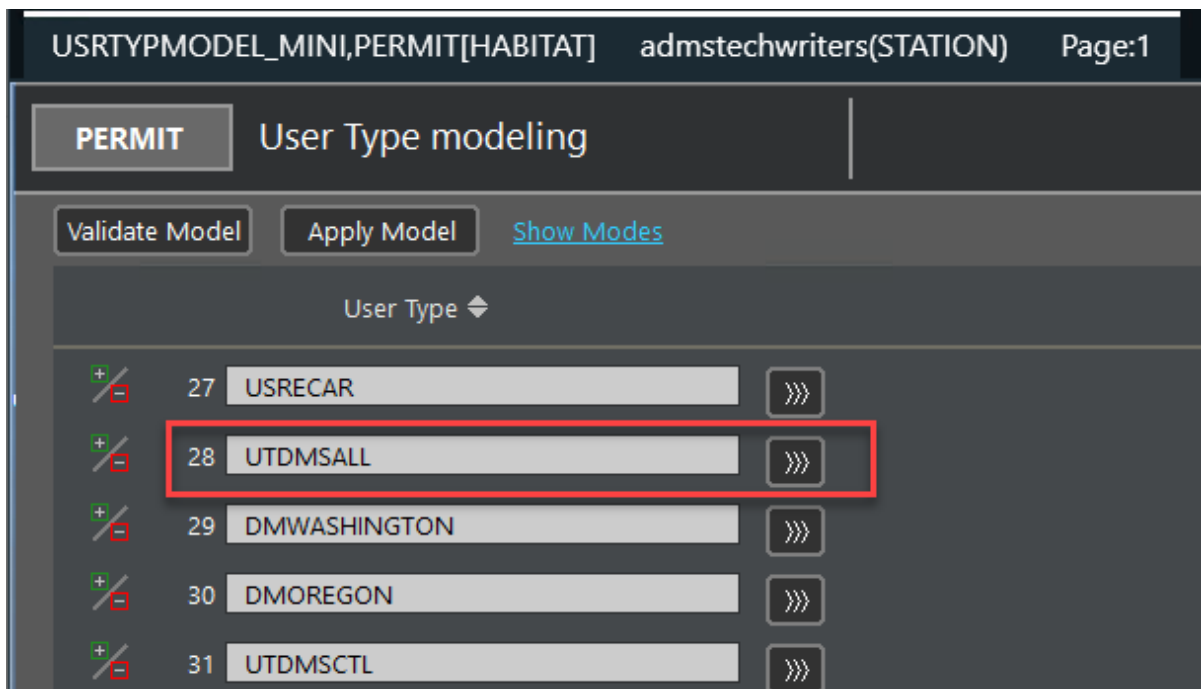
In this example, the user type is UTDMSA00.



7. Go to the User Type Modeling analyst display (Analyst Displays menu > User Modeling > User Type Modeling).

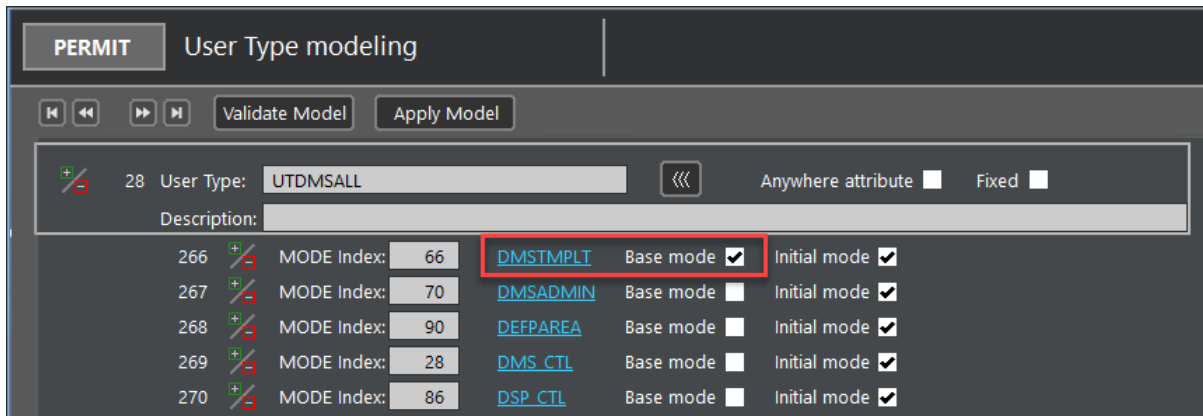


8. On the User Type Modeling display, click the expand (>>>) button for the identified user type (in this case, UTDMSALL).



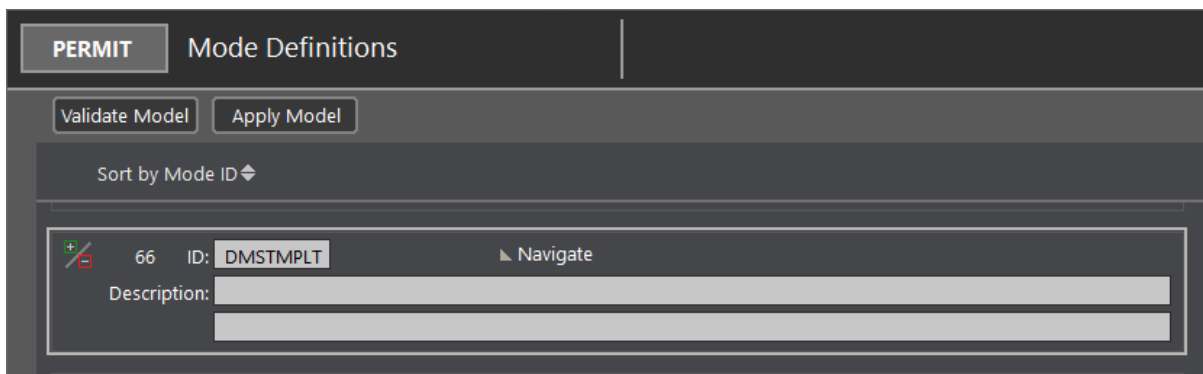
9. Identify the base mode for the user type.

In this example, the base mode is DMSTMPLT.

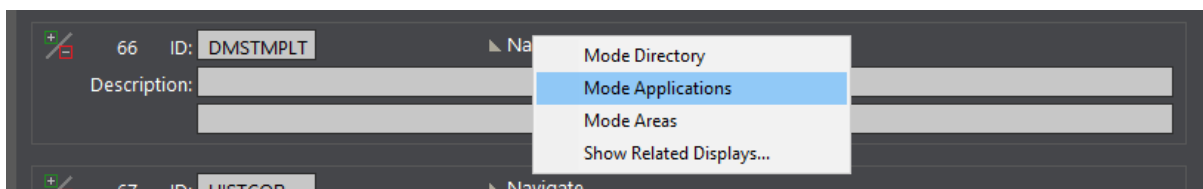


10. Click the Base Mode name.

The Mode Definitions display is shown for the base mode.



11. Right-click the Navigate field and select Mode Applications.



The Applications for Mode display is opened.

12. Scroll to where DNOM2HDB is added as a PERMIT application.

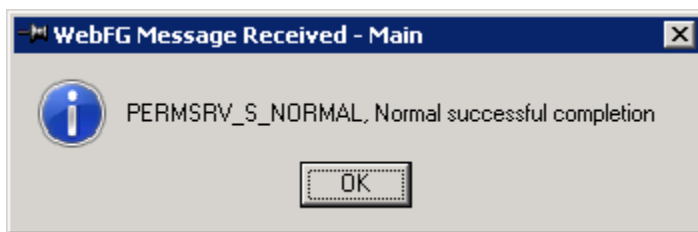
13. Enter the EMS family name, select the permission boxes, and then press Enter.

PERMIT Applications for Mode

Mode Views Validate Model Apply Model DMSTMPLT

ALARM	EMS	☒	Read	☐	Write	☐	Execute	☐	Audible
ALMPAGER		☐	Read	☐	Write	☐	Execute	☐	Audible
DNOM2HDB	EMS	☒	Read	☒	Write	☒	Execute	☒	Audible
CASESRV	HABITAT	☒	Read	☒	Write	☒	Execute	☒	Audible

14. Click Validate Model and check that the model validated successfully.

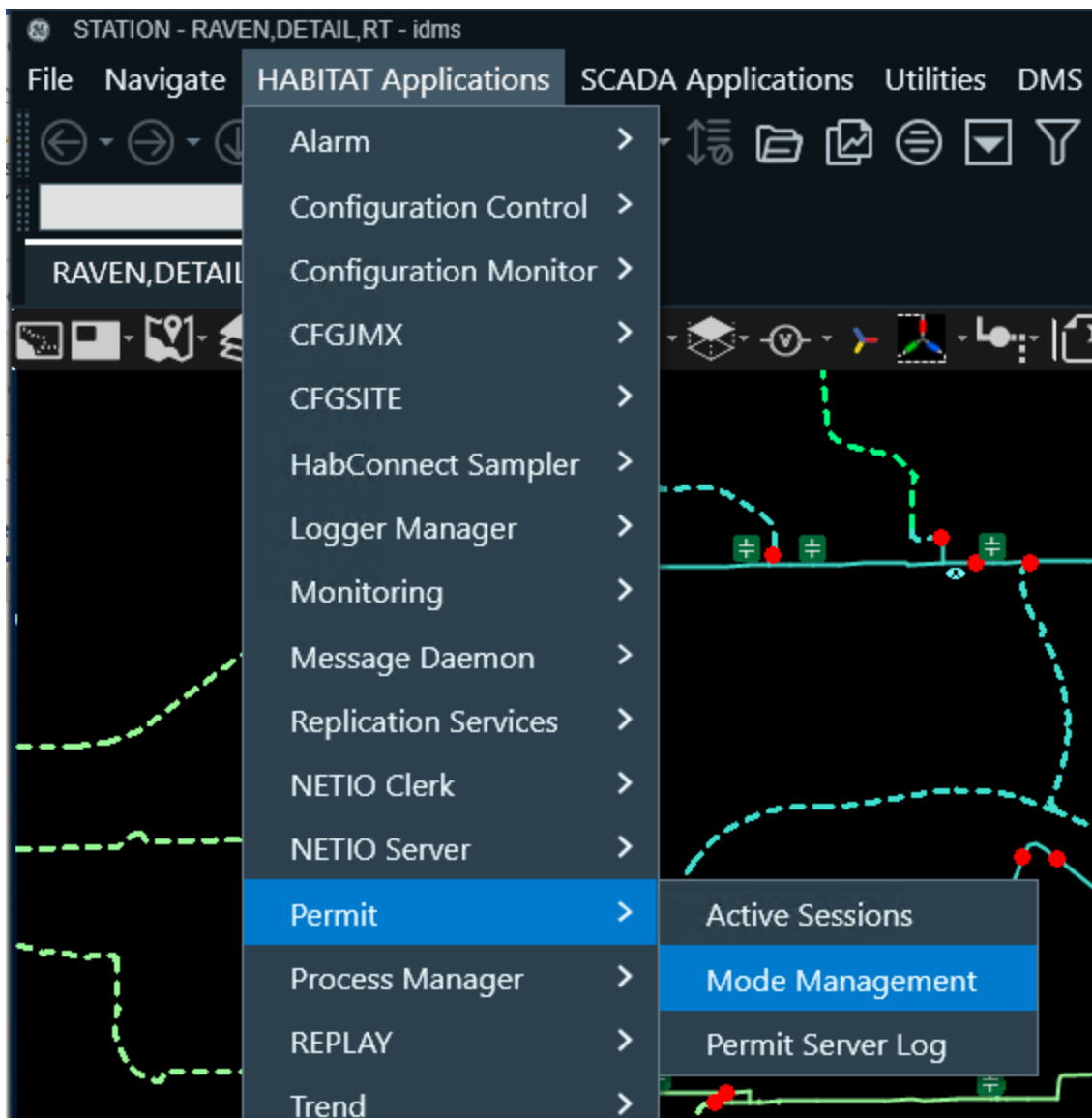


15. Click Apply Model and check that the model was applied successfully.

3.1.2. Defining the DNAFAPP Application

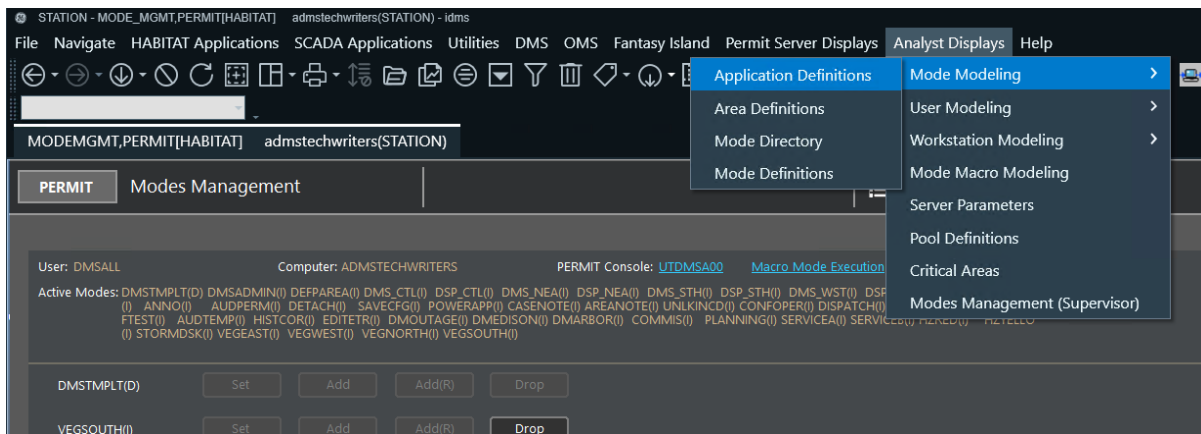
The DNAFAPP application is required to show DNAF-related DNOM2HDB displays, such as DNAF_LVMSTEP, DNAF_FISRSTEP, and DNAF_FISRPLAN.

1. In the Browser client, click HABITAT Applications menu > Permit > Mode Management.



The Mode Management display is opened for PERMIT.

2. On the Analyst Displays menu, click Mode Modeling > Application Definitions.

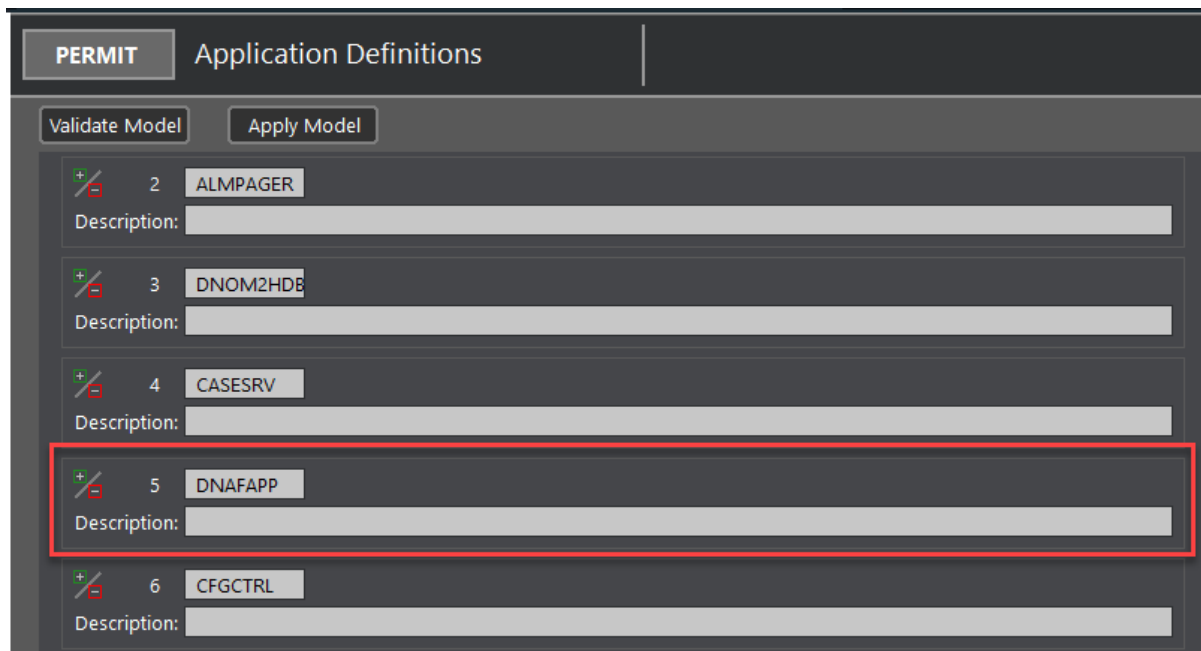


This opens the Application Definitions display.

- On the Application Definitions display, right-click on the Insert or Delete Application Record field and select the Insert Application (APP) Record option.

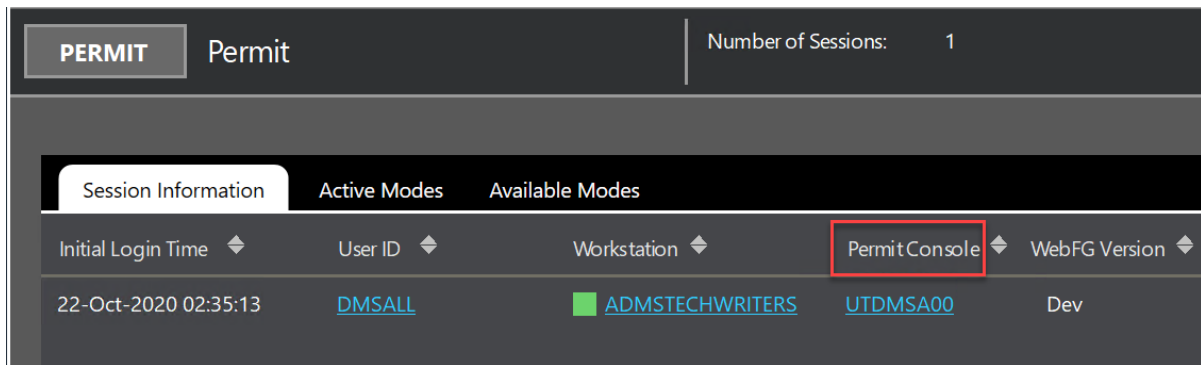
A new application record is added.

- Enter the DNAFAPP application name and press Enter.

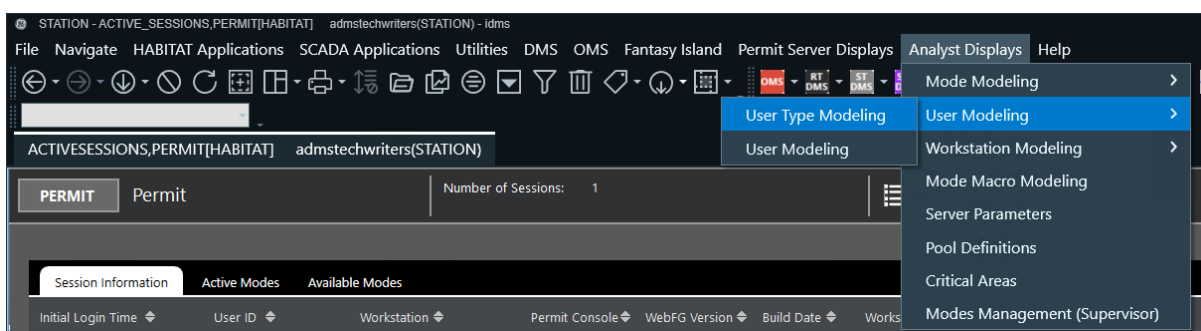


- Open the Permit main display (HABITAT Applications menu > Permit > Active Sessions).
- On the Permit main display, identify your user type on the Permit Console.

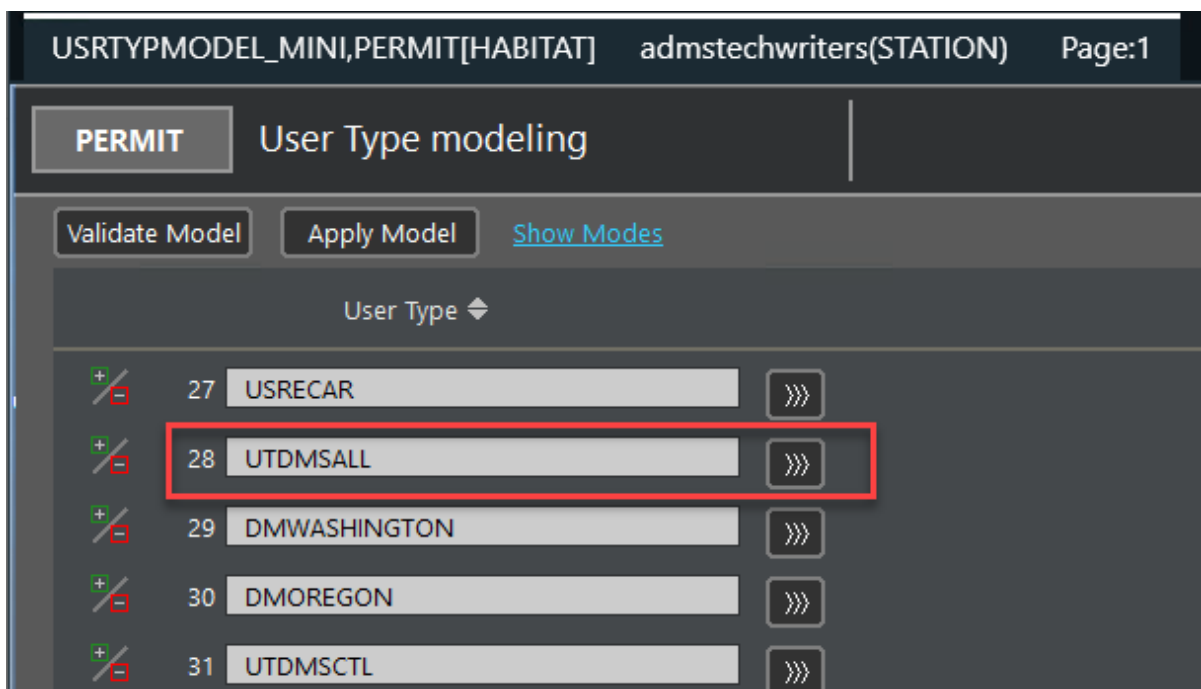
In this example, the user type is UTDMSA00.



- Go to the User Type Modeling analyst display (Analyst Displays menu > User Modeling > User Type Modeling).

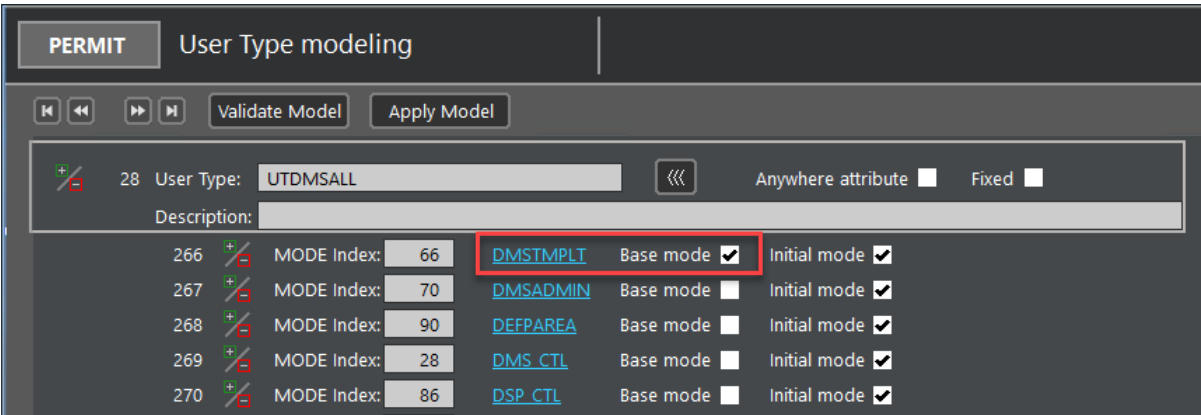


- On the User Type Modeling display, click the expand (>>>) button for the identified user type (in this case, UTDMSALL).



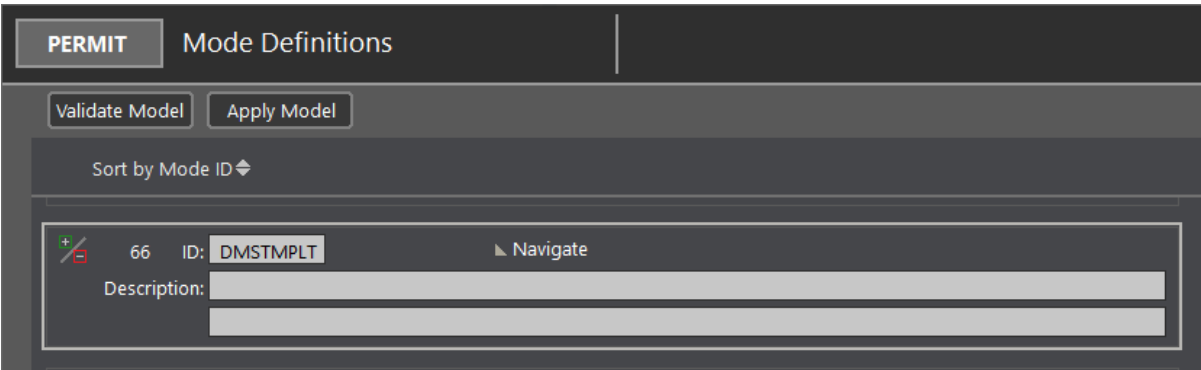
- Identify the base mode for the user type.

In this example, the base mode is DMSTMPLT.

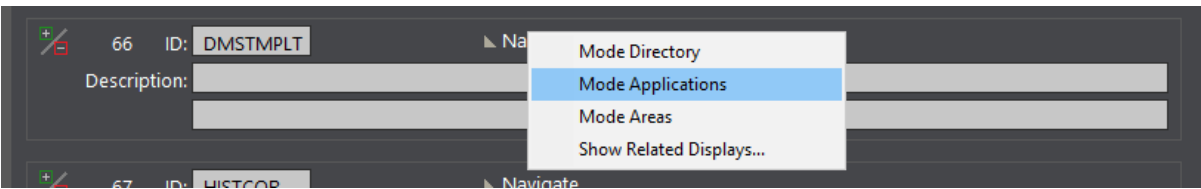


10. Click the Base Mode name.

The Mode Definitions display is shown for the base mode.



11. Right-click the Navigate field and select Mode Applications.



The Applications for Mode display is opened.

12. Scroll to where the DNAFAPP is added as a PERMIT application.

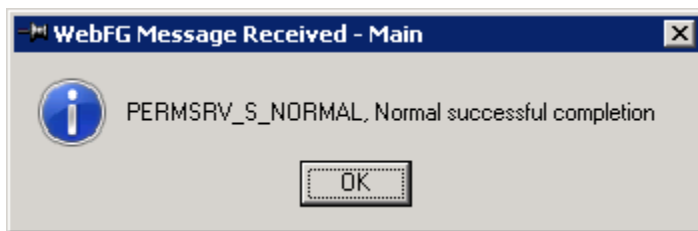
13. Enter the DNAFAM family name, select the permission boxes, and press Enter.

PERMIT Applications for Mode

Mode Views Validate Model Apply Model DMSTMPLT

ALARM	EMS	☒ Read	☐ Write	☐ Execute	☐ Audible
ALMPAGER		☐ Read	☐ Write	☐ Execute	☐ Audible
DNOM2HDB	EMS	☒ Read	☒ Write	☒ Execute	☒ Audible
CASESRV	HABITAT	☒ Read	☒ Write	☒ Execute	☒ Audible
DNAFAPP	DNAFAM	☒ Read	☒ Write	☒ Execute	☒ Audible

14. Click Validate Model and check that the model validated successfully.



15. Click Apply Model and check that the model was applied successfully.

3.2. PROCMAN

DNOM2HDB is declared as a PROCMAN task. It can be automatically started after the main PROCMAN tasks are started, or it can be manually started (in this case, do not select the Init State check box for the RUN_DNOM2HDB flag).

3.2.1. PROCMAN Model Changes

This section describes adding the DNOM2HDB task to the PROCMAN model. For more information about configuring the Habitat PROCMAN model, refer the *PROCMAN User Guide*.

! Important

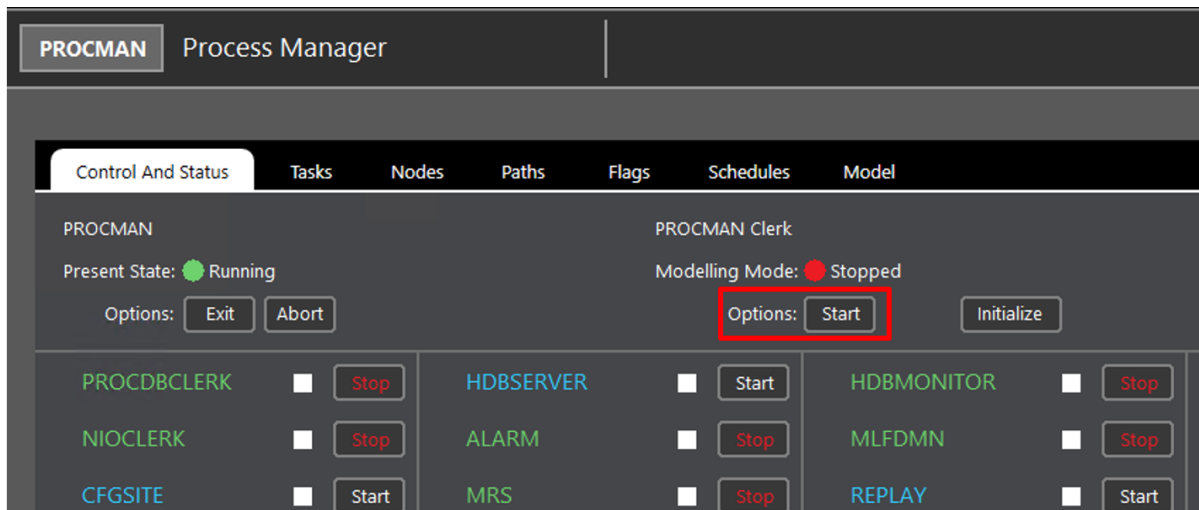
The DNOM2HDB PROCMAN task already exists in the default model and it does not need to be manually added.

To manually add the DNOM2HDB task to the PROCMAN model:

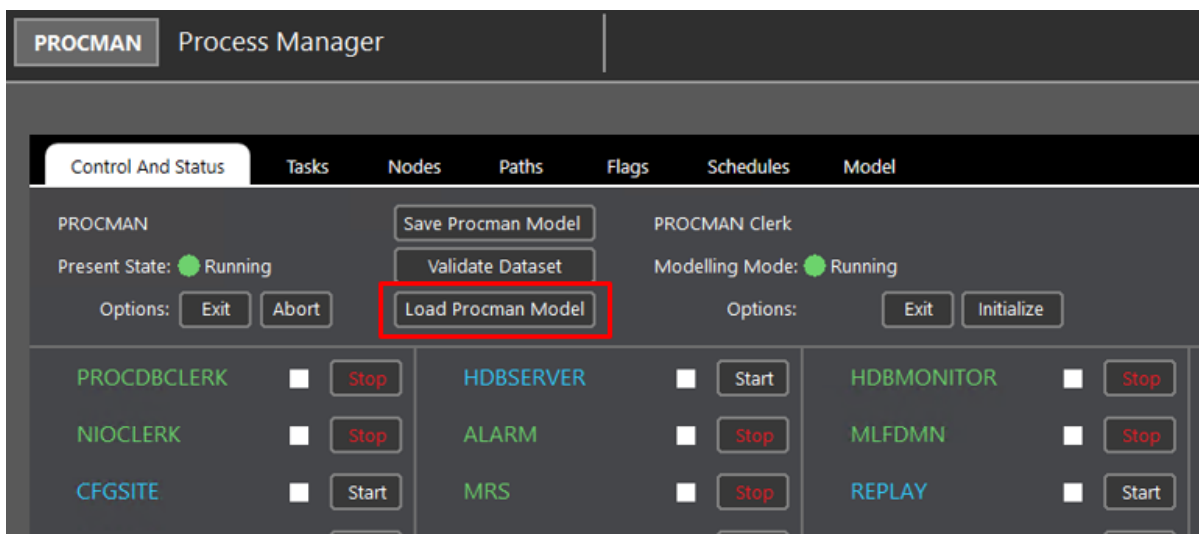
1. Stop Habitat.
2. Start Habitat without loading ADMS-specific configurations by using the command:

```
tview r procman procman habitat
```

3. Start the Browser client via the default Browser shortcut without loading any ADMS-specific configurations.
4. Log in as EMS (without password).
5. Go to the Process Manager display (HABITAT Applications menu > Process Manager > Process Manager Master).
6. Start the PROCMAN Clerk modeling mode.



7. On the Control and Status tab, click Load Procman Model.



8. On the Model Tab, add the DNOM2HDB task, flag, flag conditions, node, and path as shown in [DNOM2HDB Task Model](#) image.
9. On the Control and Status tab, select Validate Dataset to verify the model file.



10. Click Save Procman Model to save the model file.
11. Exit the PROCMAN Clerk.
12. Stop Habitat.
13. Stop the Browser client.
14. Start Habitat with ADMS-specific PROCMAN tasks.
15. Start the Browser client with ADMS-specific configurations tasks.

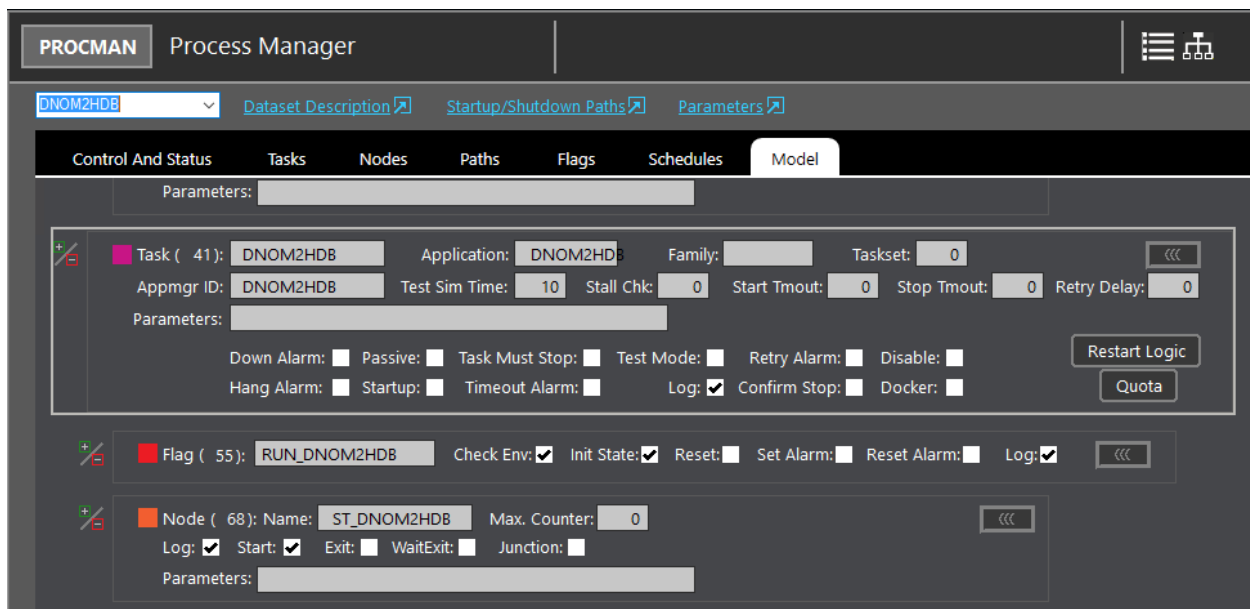


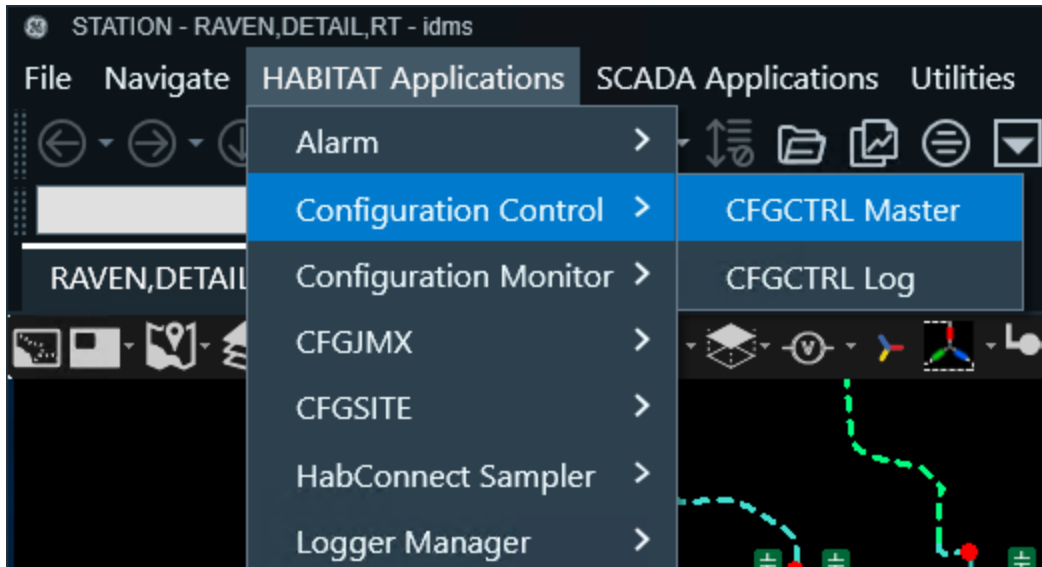
Figure 1. DNOM2HDB Task Model

3.3. Configuration Control

DNOM2HDB must be defined in the Configuration Control model. This configuration is not needed in a simulator environment.

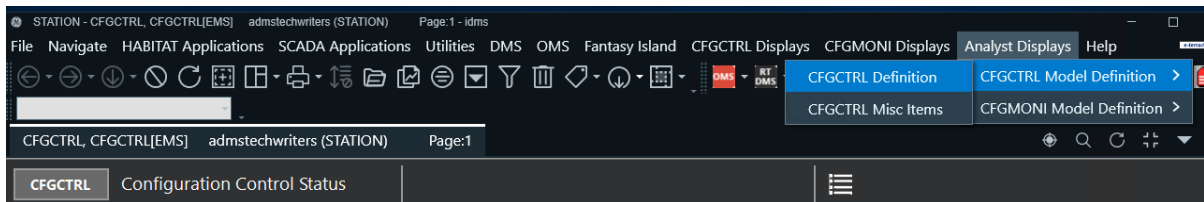
To define DNOM2HDB in the Configuration Control model:

1. Navigate to HABITAT Applications menu > Configuration Control > CFGCTRL Master.



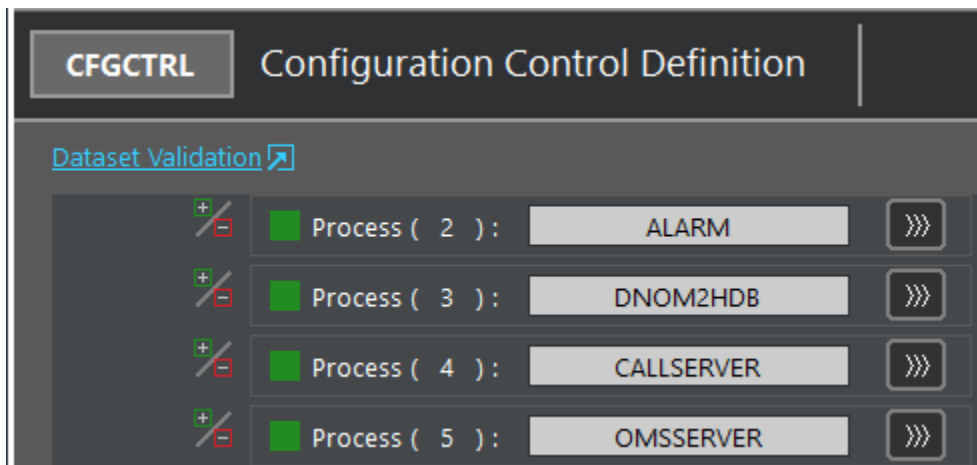
The Analyst Displays menu appears.

2. Navigate to Analyst Displays menu > CFGCTRL Model Definition > Control > CFGCTRL Definition.

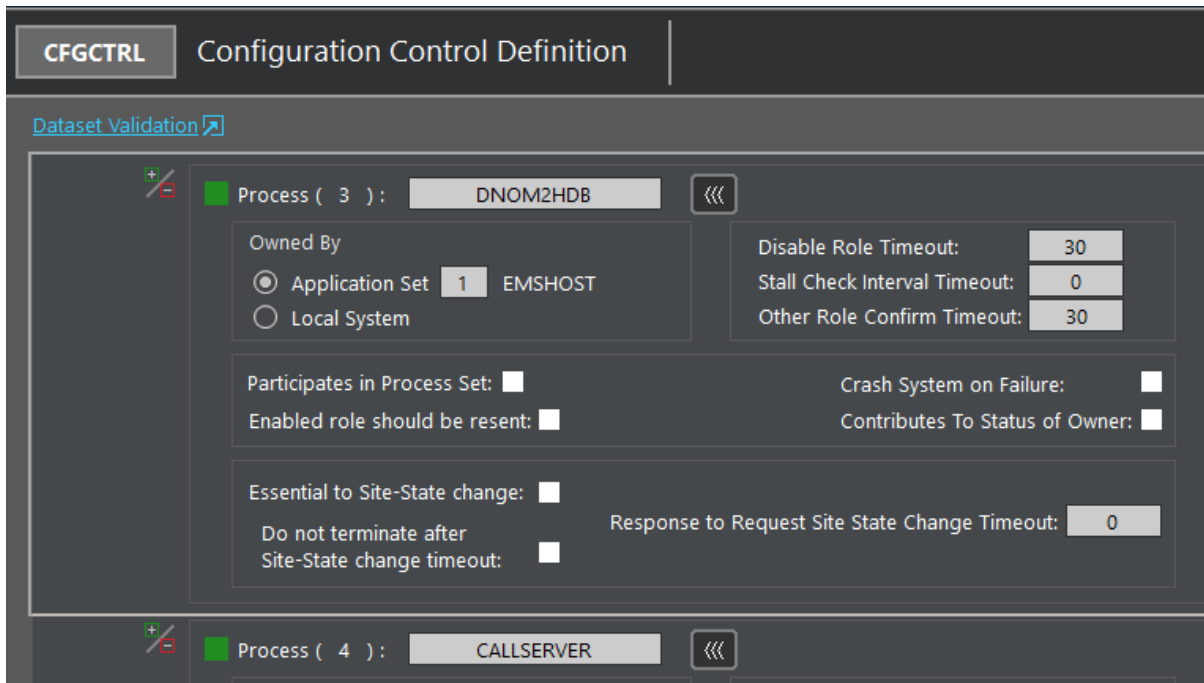


The Configuration Control Definition display opens.

3. In the Configuration Control Definition display, add a new process for DNOM2HDB.

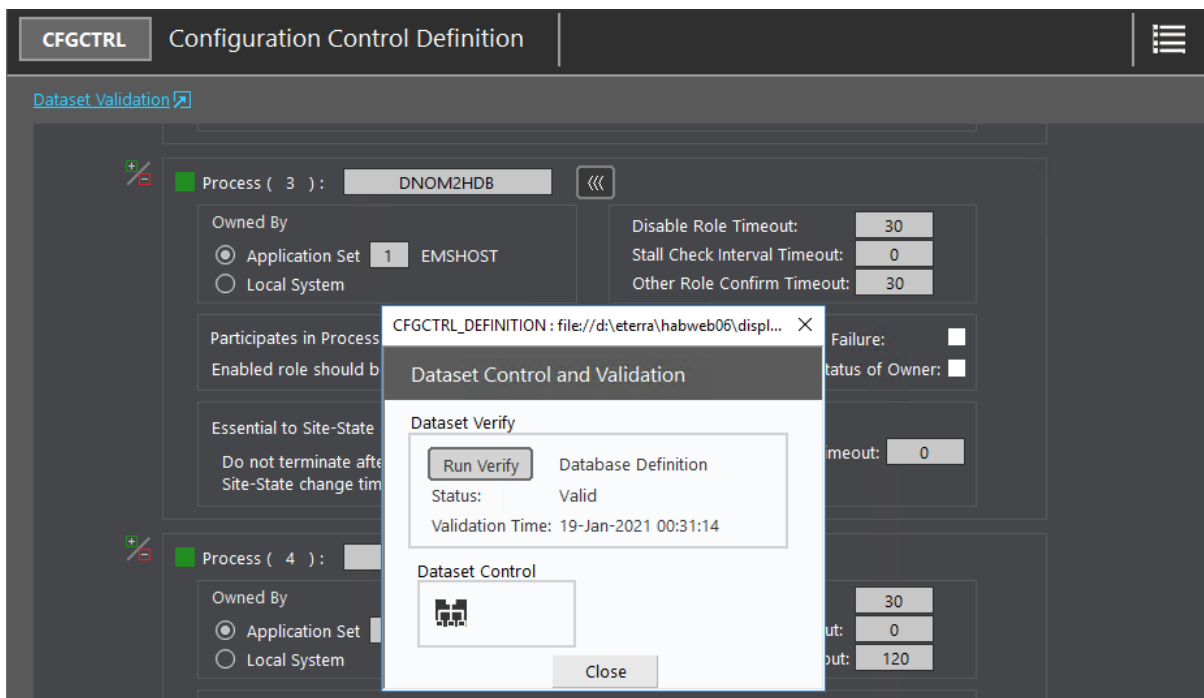


4. Click the expand button (>>>) for the DNOM2HDB process, and define the detail information for this process.

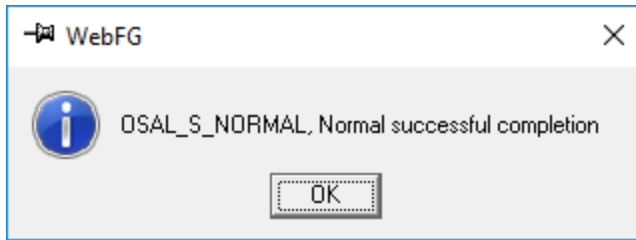


5. Navigate to HABITAT Applications menu > Process Manager > Process Manager Master.
6. Stop the CFGCTRL task.
7. Return to the Configuration Control Definition display with the DNOM2HDB process details.

The Dataset Control and Validation dialog box opens.



8. Click the Run Verify button to run validation.



9. Start the CFGCTRL task again.

3.4. Displays

Make sure that displays are stored in the following folders:

- Displays from DNOM2HDB are stored in %HABITAT_WEBROOT%\displays\dnom2hdb.
- DNAF displays (DNAF_LVMSTEP, DNAF_FISRSTEP, and DNAF_FISRPLAN) are stored in %HABITAT_WEBROOT%\displays\dnafapp.

3.5. Menus

Add the DNOM2HDB menu items to the HABITAT menu.

1. Open the webfg_menus.txt file from: %HABITAT_WEBROOT%\config\menus.
2. Go to the section that configures the HABITAT Applications menu. This menu is intended to provide high-level access to each of the HABITAT applications.

```
menu="MENU2",label="HABITAT Applications"
```

3. Search for the last menu parameter. For example:

```
parent="MENU2_21",label="License Metering Log",command="display/app=lmeter lmeter_log"
```

4. Add the menu for the DNOM2HDB items by adding the following information below the last HABITAT menu item. For example:

```
menu="MENU2_22",parent="MENU2",label="Dnom2Hdb",application="DNOM2HDB"  
parent="MENU2_22",label="Queries",command="display/app=DNOM2HDB/fam=%disfam% query_summary"  
parent="MENU2_22",label="HDB Tables",command="display/app=DNOM2HDB/fam=%disfam% table_summary"  
parent="MENU2_22",label="Parameters",command="display/app=DNOM2HDB/fam=%disfam% config"  
parent="MENU2_22",label="Stations",command="display/app=DNOM2HDB/fam=%disfam% station_summary"  
parent="MENU2_22",label="Log",command="display/app=DNOM2HDB/fam=%disfam% log"  
parent="MENU2_22",label="Debug",command="display/app=DNOM2HDB/fam=%disfam% debug"
```

```

webfg_menus.bt
308     parent="MENU2_19",label="Data Entry Message Log",command="display/app=rfgutils rfgdice_log"
309     menu="MENU2_20",parent="MENU2",label="ET Navigate",application="etnav"
310     parent="MENU2_20",label="ET Navigate Log",command="display/app=etnav etnav_log"
311     menu="MENU2_21",parent="MENU2",label="License Metering",application="lmeter"
312     parent="MENU2_21",label="License Metering Master",command="display/app=lmeter lmeter_master"
313     parent="MENU2_21",label="License Metering Log",command="display/app=lmeter lmeter_log"
314     menu="MENU2_22",parent="MENU2",label="Dnom2Hdb",application="DNOM2HDB"
315     parent="MENU2_22",label="Queries",command="display/app=DNOM2HDB/fam=%disfam% query_summary"
316     parent="MENU2_22",label="HDB Tables",command="display/app=DNOM2HDB/fam=%disfam% table_summary"
317     parent="MENU2_22",label="Parameters",command="display/app=DNOM2HDB/fam=%disfam% config"
318     parent="MENU2_22",label="Stations",command="display/app=DNOM2HDB/fam=%disfam% station_summary"
319     parent="MENU2_22",label="Log",command="display/app=DNOM2HDB/fam=%disfam% log"
320     parent="MENU2_22",label="Debug",command="display/app=DNOM2HDB/fam=%disfam% debug"
321

```

5. Save your changes.
6. Restart the Browser client.

The DNOM2HDB menu is now added to the HABITAT Applications menu.

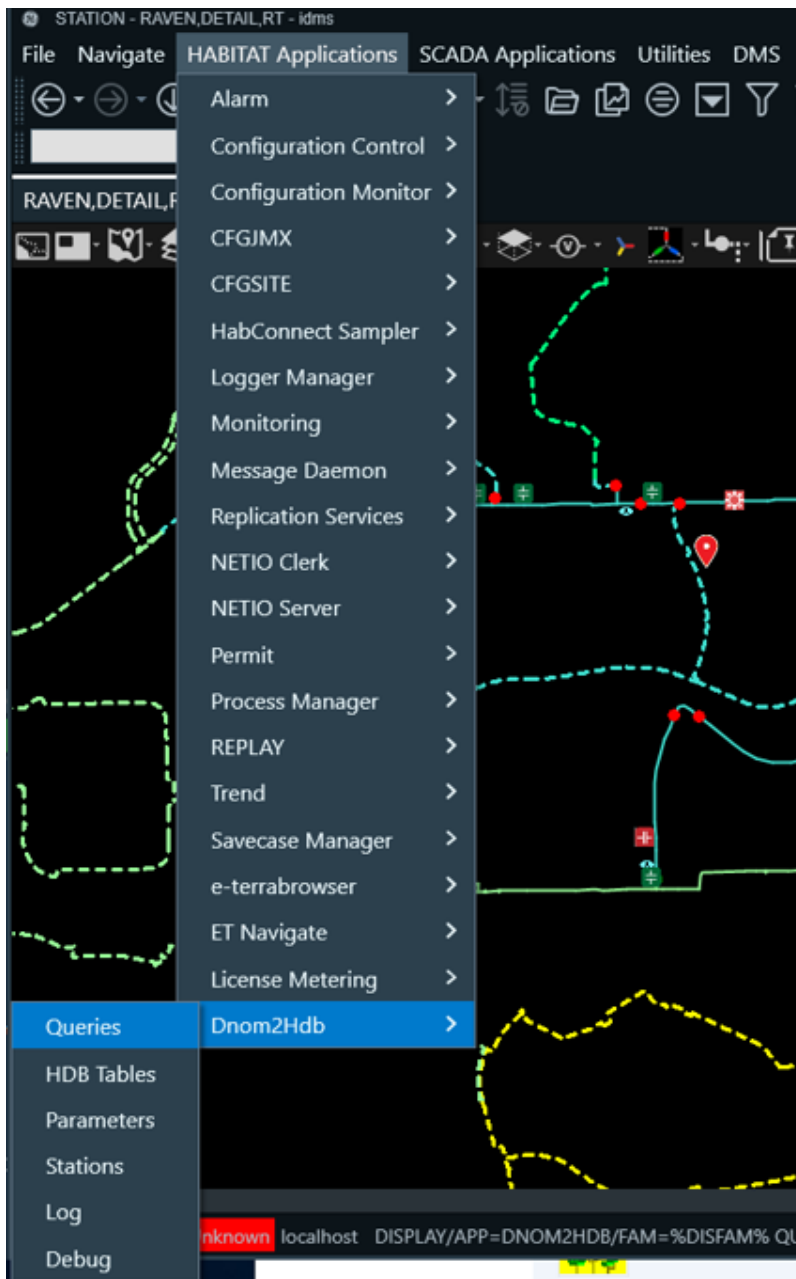


Figure 2. DNOM2HDB Menu

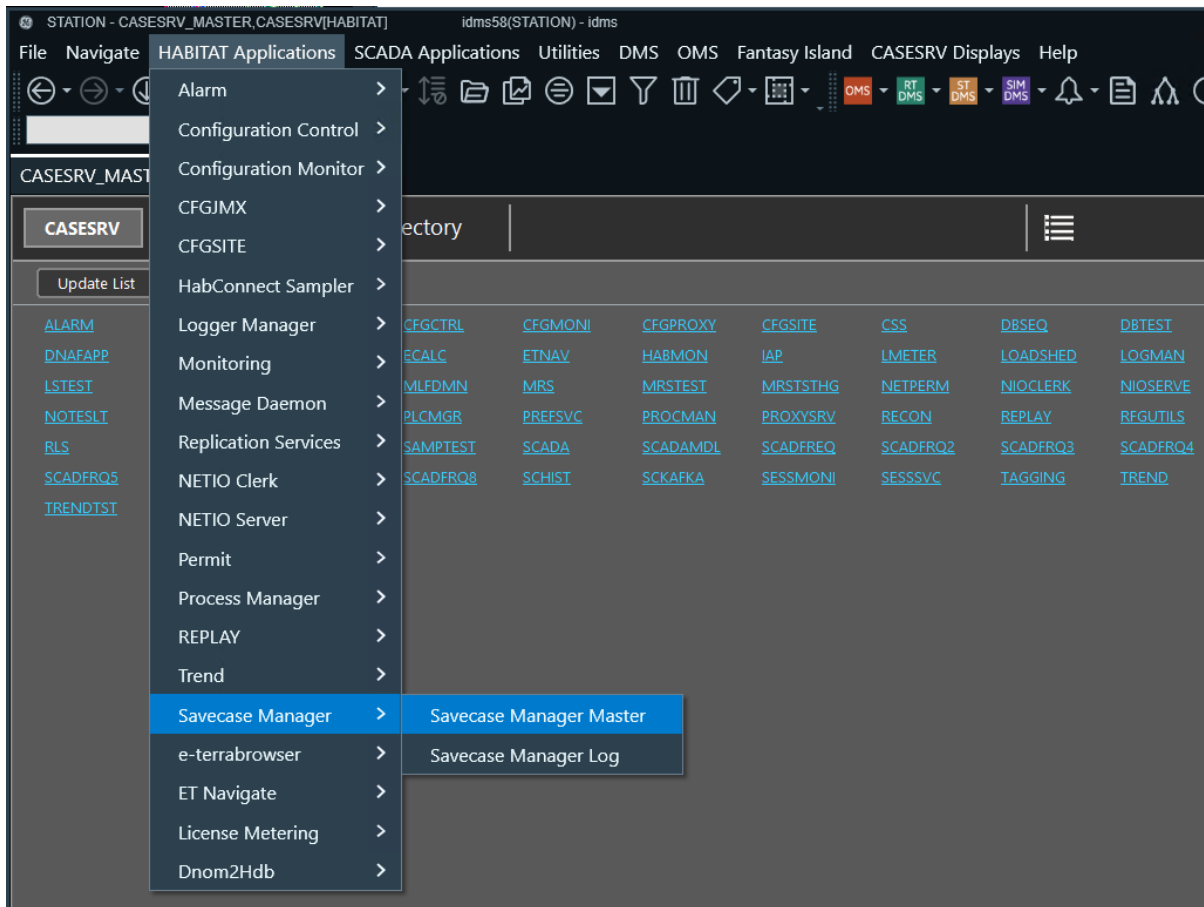
3.6. Savecases

DNOM2HDB has an initial savecase to set up the default values.

To retrieve the initial savecase:

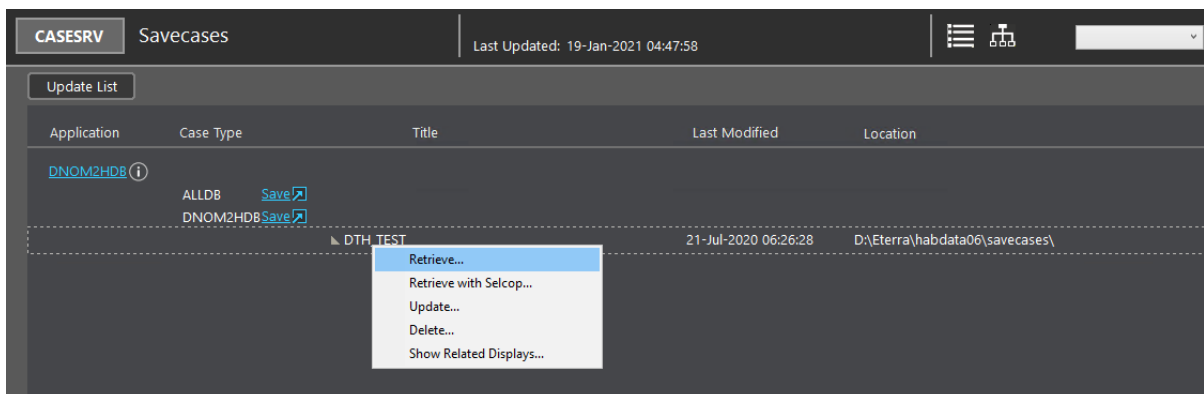
1. Navigate to HABITAT Applications menu > Process Manager > Process Manager Master.
2. Stop the DNOM2HDB task by clicking the Stop button.

3. Navigate to HABITAT Applications menu > Savecase Manager > Savecase Manager Master.

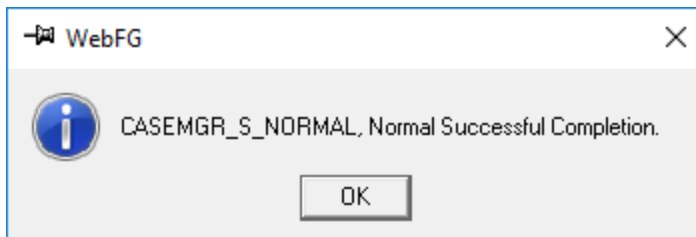
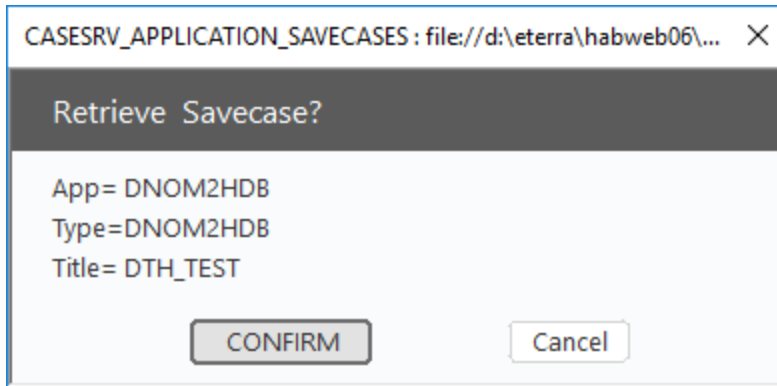


This opens the Savecase Application Directory display.

4. Select the DNOM2HDB task.
5. Right-click on the title and select Retrieve.



6. Click Confirm to retrieve the savecase.



7. Navigate to HABITAT Applications menu > Process Manager > Process Manager Master.
8. Start the DNOM2HDB task.

3.7. DMS Service Module

To establish a connection between the DNOM2HDB application and the DMS Service module on the DNAF server:

1. Open the DNAF server configuration file `NetServerDnafConfig.txt` (stored in `%IDMS_ROOT%\config\server`) in a text editor.
2. Find the "Service module listening socket and socket parameters" section, and uncomment the following parameters:

```
SERVICE_PORT 6033
SERVICE_SOCKET_OUTPUTBUF_MAX_SIZE 6000000
SERVICE_SOCKET_WOULDBLOCK_RETRY 2
SERVICE_SOCKET_MAX_CLIENT_MES_SIZE 640000
```

3. Make sure that `SERVICE_PORT` is assigned to the port number available on your server.
4. Restart the DNAF server.
5. Open the DNOM2HDB parameters display (HABITAT menu > DNOM2HDB > Parameters).
6. Make sure the Host Name and Socket port of your DMS Service module on the DNAF servers match the DNOM2HDB parameter settings.

The screenshot shows the 'Parameters' tab of the DNOM2HDB configuration window. The window title is 'B - CONFIG,DNOM2HDB[EMS] idms58 (B) - idms'. The menu bar includes File, Navigate, HABITAT Applications, SCADA Applications, Utilities, DMS, OMS, Fantasy Island, DnomToHdb, and Help. The toolbar contains various icons for navigation and actions. The main content area is divided into sections for configuring different components.

Parameters ☐ Freeze heartbeat **DMS Up** 19-Jan-2021 05:27:40

Heartbeat Seconds

DMS Server #1:

Host Name:

Socket:

Timeout: Seconds

DMS Server #2:

Host Name:

Socket:

Timeout: Seconds

Query menu for DNAF applications

	Label Text	DNAF Status
DPF	DPF	Solved / BLA Converged
FISR	FISR	Solved
LVM	LVM	Solved

Note

The DNOM2HDB process must be started to set the parameters.

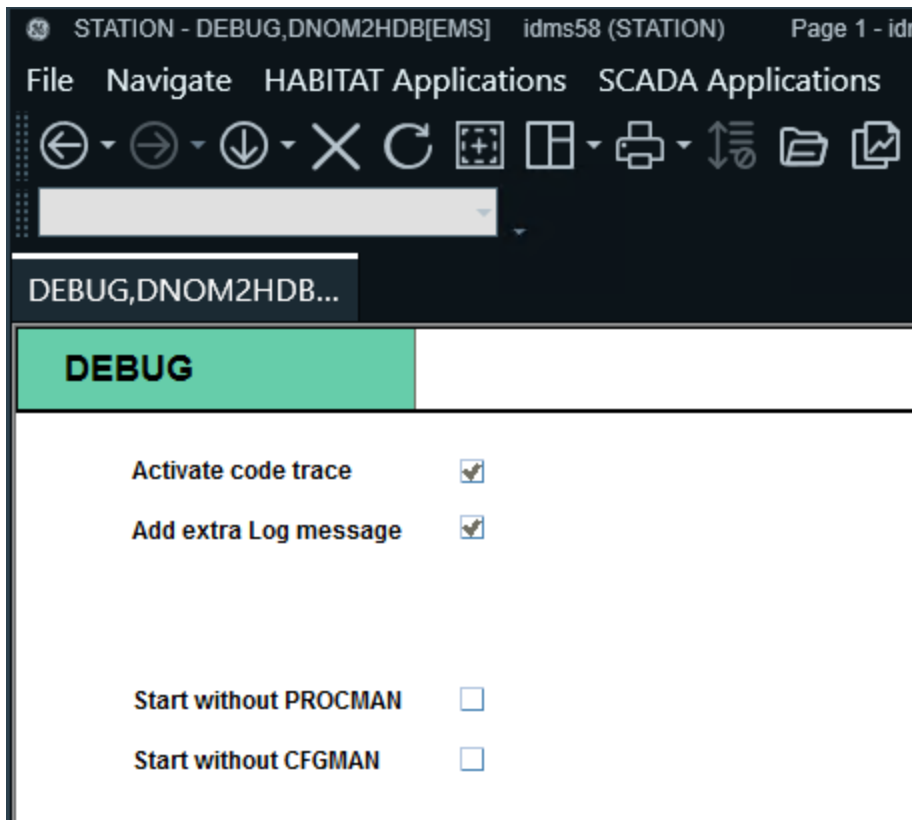
3.8. Troubleshooting

3.8.1. Enabling Detailed Logging

Some extra tracing can be logged in the Log display or to an ASCII output file by selecting the Activate Code Trace option from the Debug display or by defining the SAF_CODE_DEBUG environment variable.

To enable detailed logging from the DNOM2HDB Debug display:

1. Open the DNOM2HDB Debug display.



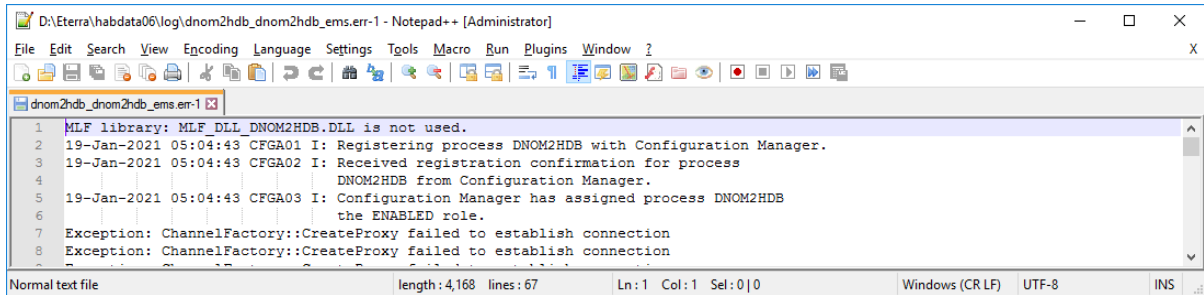
2. Select the Activate Code Trace and Add Extra Log Message options.
3. Navigate to the %HABITAT_LOGDIR% folder (D:\Eterra\habdata06\log).
4. Open a dnom2hdb.out file. This file contains errors and detailed responses for queries that are performed by the DNOM2HDB application.

```

1  _profiler_ Tracing dmsframework Enabled: NO
2  SPYCODE trace is ON
3  ----> ProcmanCfgmanRegisterTask
4  Go to PpmMainLoop...
5  --> CfgmanRegister_cb
6  CFGMAN -> CFGMAN_ROLE_ENABLED
7  SPYCODE trace is ON
8  DEBUG: Tue Jan 19 07:02:38 2021
9  threadId: b8Start Method Invocation connect(); ,clientId = 4000 ,requestId = 3000
10 INFO: Tue Jan 19 07:02:38 2021
11 threadId: 299connection established, clientId = 4000
12
13 CommunicationEventsCallbackImpl::OnConnectEvent() host: localhost port: 6033
14 DEBUG: Tue Jan 19 07:02:38 2021
15 threadId: b8Successfully Finish Method Invocation (); ,clientId = 4000 ,requestId = 3000 method name = connect
16 DEBUG: Tue Jan 19 07:02:38 2021
17 threadId: b8Start Method Invocation Subscribe(); ,clientId = 4000 ,requestId = 3001
18 DEBUG: Tue Jan 19 07:02:38 2021
19 threadId: b8Successfully Finish Method Invocation (); ,clientId = 4000 ,requestId = 3001 method name = Subscribe
20 DEBUG: Tue Jan 19 07:02:39 2021
21 threadId: b8Start Method Invocation ExecuteQuery(string SELECT NAME, AFRSTATUS, DPFSTATUS, FISRSTATUS, FLSTATUS, VVCSTATUS,
22 DEBUG: Tue Jan 19 07:02:39 2021
23 threadId: b8Successfully Finish Method Invocation (string <dnom><SUB><NAME>3LIONS</NAME>
24 <AFRSTATUS>Unsolved</AFRSTATUS>
25 <DPFSTATUS>Solving</DPFSTATUS>
26 <FISRSTATUS>Unsolved</FISRSTATUS>

```

5. Open a `dnom2hdb.err` file. This file contains application Configuration Manager and Process Manager statuses and errors.



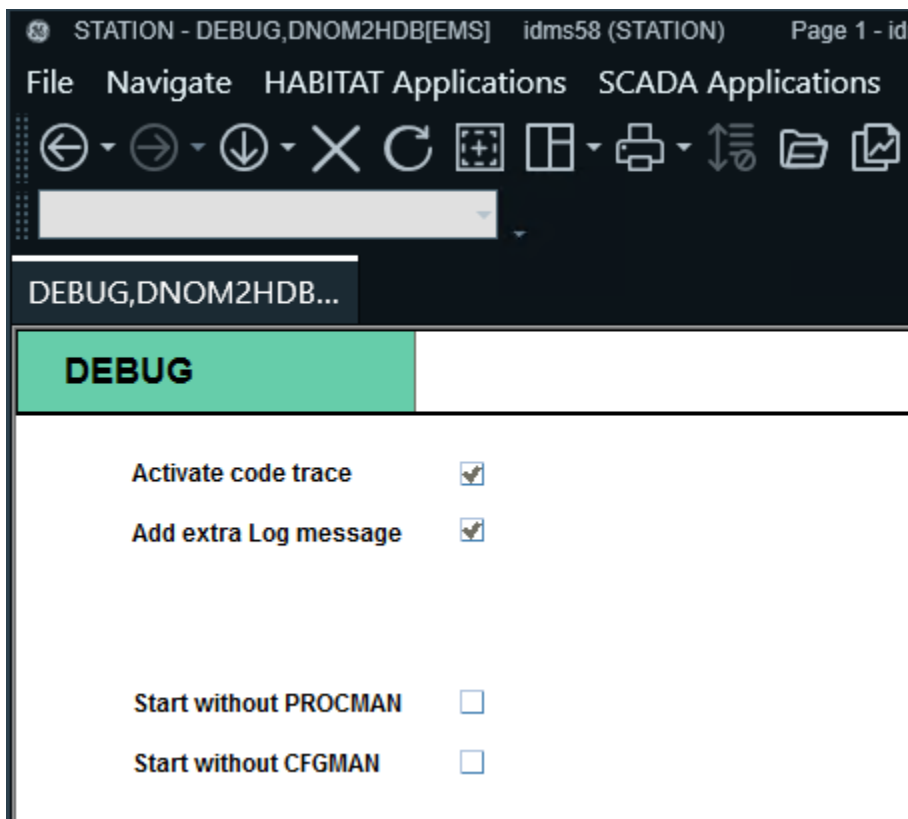
```
D:\Eterra\habdata06\log\dnom2hdb_dnom2hdb_err-1 - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
dnom2hdb_dnom2hdb_err-1
1 MLF library: MLF_DLL_DNOM2HDB.DLL is not used.
2 19-Jan-2021 05:04:43 CFMGA01 I: Registering process DNOM2HDB with Configuration Manager.
3 19-Jan-2021 05:04:43 CFMGA02 I: Received registration confirmation for process
4 DNOM2HDB from Configuration Manager.
5 19-Jan-2021 05:04:43 CFMGA03 I: Configuration Manager has assigned process DNOM2HDB
6 the ENABLED role.
7 Exception: ChannelFactory::CreateProxy failed to establish connection
8 Exception: ChannelFactory::CreateProxy failed to establish connection
Normal text file length: 4,168 lines: 67 Ln: 1 Col: 1 Sel: 0|0 Windows (CR LF) UTF-8 INS
```

3.8.2. Running DNOM2HDB without PROCMAN

For quick installations, it is possible to run DNOM2HDB even if models of PROCMAN and CFGMAN are not updated.

To run DNOM2HDB without updating the PROCMAN and CFGMAN models:

1. Open the DNOM2HDB Debug display.



2. Select the Start without PROCMAN and Start without CFGMAN options.